

Ethnopharmacology, LC-MS profiling, and molecular docking of diarrheal-healing plants by the Osing tribe

Khrisna Agung Cendekiawan^{1,3}, Kintoko^{1,2*}, Sapto Yuliani¹, Firdha Aprillia Wardhani⁴

¹ Doctoral Pharmacy Program, Faculty of Pharmacy, Ahmad Dahlan University, Indonesia

² Department of Pharmaceutical Biology, Faculty of Pharmacy, Universitas Ahmad Dahlan, Indonesia

³ Bachelor of Pharmacy Program, Faculty of Health Sciences, Universitas dr. Soebandi, Indonesia

⁴ Bachelor of Pharmacy Program, Faculty of Pharmacy, Universitas Jember, Indonesia

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ABSTRACT

Indonesia has a long history of using medicinal plants, but much of this knowledge, especially among groups like the Osing people in East Java, is not well-documented. This study looks at the plants used by the Osing community to treat diarrhea. This research was conducted through interviews with residents of Banjar and Licin Villages in Banyuwangi Regency. The significance and perceived efficacy of each medicinal plant were evaluated using the ethnobotanical indices Use Value (UV) and Fidelity Level (FL). They also did tests to find the active chemicals in the plants. The most common plants for treating diarrhea were *Psidium guajava* (UV = 2.44) and *Piper betle* (UV = 1.11). *Psidium guajava* had the highest FL value (64.7%). Tests have shown that these plants contain essential chemicals. Further analysis confirmed the presence of quercetin and kaempferol, which are known to help alleviate stomach problems. To support these findings, computer-based tests were done, including Lipinski rule prediction, ADMET profiling, and molecular docking. The results showed quercetin binds well ($\Delta G = -9.72$) and has a good RMSD value (1.83 Å). This suggests quercetin could be a good starting point for making antidiarrheal drugs. Overall, the study underscores the significance of traditional knowledge and recommends further research on plant-based treatments for stomach issues.

Keywords: Ethnopharmacology; Anti diarrheal; Phytochemical; Molecular Docking

Corresponding author:

Kintoko,

Universitas Ahmad Dahlan

Prof. Dr. Soepomo, S.H. Street, Umbulharjo, Yogyakarta, Indonesia

Email: kintoko@pharm.uad.ac.id



INTRODUCTION

Medicinal plants play a significant role in Indonesian culture and are a vital component of traditional healthcare. People have used these plants for healing for many generations (Arozal et al., 2020). The Osing community in East Java, located in the Banyuwangi Regency near the Ijen Mountains, is especially interesting for its use of these plants (Prasetyo et al., 2018). While there is a wealth of information about the Osing people's religious and ritual practices, there is relatively little written about their use of medicinal plants. The Osing use different treatment methods compared to other tribes, focusing on both the environment and the healing properties of plants. They reside in the Ijen National Park area and employ specific herbal combinations, such as *Psidium guajava* and *Piper betle*, which have been well-tested and hold strong cultural and healing significance (Kusumo et al., 2023).

Ethnopharmacology is a method for documenting and studying traditional medicine across cultures. It combines many fields to understand how traditional remedies work scientifically (Sumarni et al., 2019). In this study, phytochemical screening was conducted to detect the presence of key metabolite classes, including carbohydrates, phenolics, carotenoids, coumarins, steroids, tannins, alkaloids, terpenoids, and saponins. Qualitative methods were used to identify both primary and secondary metabolites. Primary metabolites, though essential for plant growth and development, are typically produced in limited quantities. In contrast, secondary metabolites serve ecological roles such as defense against herbivores and pathogens, or adaptation to environmental stress. These compounds may also exhibit pharmacological properties that can be harnessed in drug discovery (Farag et al., 2020).

To complement the screening, liquid chromatography-mass spectrometry (LC-MS) was used as a sensitive and precise analytical tool to identify and quantify the specific secondary metabolites. LC-MS combines the separation capabilities of liquid chromatography with the detection and structural elucidation of mass spectrometry, enabling comprehensive analysis of complex plant extracts. This method is widely used in pharmaceutical analysis due to its high sensitivity, specificity, and ability to detect trace levels of compounds (Korfmacher, 2005).

This study looks at how the Osing people use plants to treat diarrhea. It aims to preserve their traditional knowledge and prove it scientifically. The research combines interviews with chemical tests to find active ingredients in plants. The study focuses on betel (*Piper betle*) and guava (*Psidium guajava*) plants. It uses LC-MS to identify active compounds in these plants. The study also uses molecular docking to see how these compounds form hydrogen bonds.

MATERIALS AND METHOD

Materials

Fresh leaves of *Psidium guajava* L. (Myrtaceae) and *Piper betle* L. (Piperaceae) were collected from Banjar and Licin Villages, Banyuwangi Regency, East Java, based on their ethnomedicinal use among the Osing community. The plants were authenticated using the World Flora Online database. Dried and powdered leaves were extracted with 70% methanol (Merck, Germany) by maceration for 72 hours at room temperature with occasional stirring. The filtrates were concentrated using a rotary evaporator (Heidolph, Germany) and stored at 4 °C until use. Phytochemical screening was conducted using standard protocols, employing reagents such as Dragendorff's, Mayer's, and Bouchardat's for alkaloids; magnesium-HCl for flavonoids; FeCl₃ and lead acetate for tannins; the foam test for saponins; and the Liebermann-Burchard reaction for steroids and triterpenoids. All reagents were of analytical grade and obtained from Sigma-Aldrich.

Secondary metabolite profiling was conducted using an LC-MS system (ACQUITY UPLC® H-Class, Waters, USA) with a BEH C18 column (2.1 × 50 mm) operated in positive ESI mode (*m/z* 50–1200). Chromatographic peaks were processed using MassLynx software, which identified quercetin and kaempferol as the major constituents. Identified compounds were optimized in MarvinSketch (ChemAxon) and evaluated for drug-likeness via Lipinski's Rule of

Five. Protein targets related to diarrheal mechanisms were retrieved from the Protein Data Bank, and molecular docking was performed using AutoDock and AutoDockTools 1.5.7, with interactions visualized in Molegro Molecular Viewer (MMV). Molecular dynamics simulations (100 ns) were conducted to assess complex stability and binding free energy using the MMPBSA method. All materials and computational tools were validated prior to use, and the study was approved by the Health Research Ethics Commission of Universitas Dr. Soebandi (Approval No. 569/KEPK/XI/2024).

Methods

Ethics and legal considerations

This study received ethical approval from the Health Research Ethics Commission of the Faculty of Health Sciences at Universitas Dr. Soebandi Jember, Indonesia (Approval No. 569/KEPK/XI/2024). The research adhered to the seven core ethical principles promoted by the World Health Organization, which emphasize the scientific and social value of the study, fair distribution of risks and benefits, risk reduction, avoidance of exploitation, and the safeguarding of participants' privacy, confidentiality, and autonomy through informed consent. Written informed consent was obtained from the village leadership, and verbal consent was secured from all individual participants before data collection.

Background of the study area

To enhance data representativeness, fieldwork was conducted in two principal Osing villages within the Ijen National Park: Licin and Banjar, as illustrated in Figure 1. These sites were intentionally chosen based on their deep-rooted commitment to Osing cultural practices and their ongoing engagement in the use and preservation of traditional ethnomedicinal knowledge.

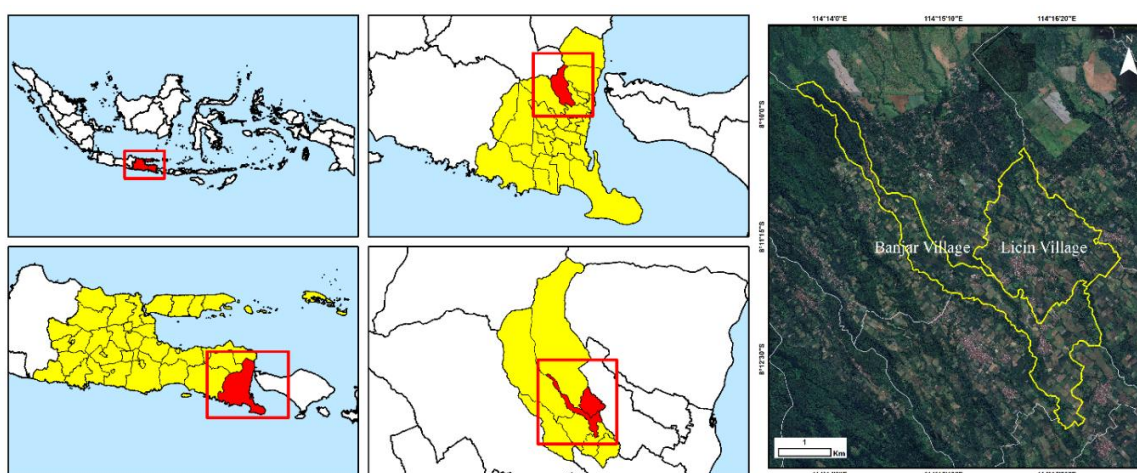


Figure 1. The study area map shows the location of Osing District within East Java Province, Banyuwangi Regency, Licin Subdistrict, Banjar Village and Licin village

Licin Village is in the Banyuwangi District. It is located at $8^{\circ}11'37.174''\text{S}$ and $114^{\circ}16'3.445''\text{E}$ and covers about 4.67 km². The village is more than 785 meters above sea level, making it one of the highest places where people live on Java Island (BPS-Statistics of Banyuwangi District, 2024). In the same district, Banjar Village is at $8^{\circ}13'0.48000''\text{S}$ and $114^{\circ}16'15.24000''\text{E}$, with an area of about 9.00 km². Banjar is the only official Osing Village in Banyuwangi District and is known for keeping traditional healing practices alive (BPS, 2024).

Ijen National Park has a unique mountain ecosystem that affects nearby villages. This highland area has a tropical mountain climate. Temperatures range from 7°C to 18°C, with humidity and clear wet and dry seasons. These conditions help many medicinal plants grow, like mountain herbs and trees essential to Osing traditional medicine. The area has forests, grasslands,

and farm terraces. People grow medicinal plants with crops like cabbage and potatoes. The volcanic soil is very fertile, helping plants with healing properties thrive. The landscape has steep hills, deep valleys, and volcanic ridges, creating a rich environment with many unique plants. This variety is seen in the Osing community's knowledge of plants. Their traditional medicine is closely linked to nature, based on generations of understanding and careful harvesting (Chandra, 2022).

Ethnopharmacology study

A study on traditional medicine was done in Banjar Village, Banyuwangi, from November to December 2024. Researchers used methods such as observation, interviews, and documentation of local medicinal practices. They chose 34 people from the Osing ethnic group who were 40 years or older, lived there for a long time, and knew about traditional herbal medicine. These people included healers, elders, and others with herbal knowledge. The study used questionnaires and interviews to gather information. Although the study does not represent all Osing people, it provides important and relevant insights into their traditional medicine, a common focus in such research.

Researchers used a snowball sampling method to find participants. They started with recommendations from respected local figures, such as traditional healers and midwives, known for their knowledge of traditional medicine. To make sure all participants were from the Osing ethnic group, they used several checks. First, they reviewed village records and received recommendations from local leaders, including the head of Banjar Village and traditional authorities who know community family histories. Participants also had to speak Osing, know Osing customs and rituals, and take part in traditional ceremonies or medicinal practices. These cultural signs, along with living in Banjar or Licin Villages, confirmed their Osing identity. This careful process ensured only actual Osing community members, recognized for their traditional medicinal knowledge, were included in the study (Jadid et al., 2020; Nugraha et al., 2024). During fieldwork, researchers collected plant samples. They identified these plants based on observations and participant information. They also used existing literature on traditional medicine in the area. To confirm the plant types, they checked them against the World Flora Online database (<https://wfoplantlist.org/>).

Quantitative analysis of ethnomedicinal data

To evaluate traditional medicinal knowledge, a series of ethnobotanical indices were utilized, specifically the Use Value and Fidelity Level. These analytical tools facilitated the assessment of the relative indigenous knowledge and the degree of consensus regarding the utilization of medicinal plant species within the target community.

Use value (UV)

The Use Value (UV) index (1) was used to quantify the relative importance of each medicinal plant species reported by respondents for the treatment of diarrhea and pain. This metric identifies plant species with notable therapeutic relevance and broad cultural use (Ralte et al., 2024).

$$\text{Value (UV)} = \sum U / N \dots\dots\dots (1)$$

Where $\sum U$ = number of use-reports cited for a species by all respondents, N = total number of respondents.

In this context, UV denotes the Use Value of a given plant species, where U represents the total number of citations or mentions by respondents, and N signifies the overall number of participants interviewed. A higher UV score reflects frequent citation, indicating greater cultural and medicinal relevance, whereas a UV approaching zero implies minimal usage and limited significance within the local ethnopharmacological framework.

Fidelity level (FL)

The Fidelity Level (FL) (2) index assesses the degree of specificity with which a particular plant species is used to address specific health conditions, such as diarrhea and pain. Derived from the level of agreement among respondents, this metric illustrates the extent to which a given species is preferentially employed for a defined therapeutic application (Leonti, 2022).

$$\text{Fidelity Level (FL)} = (N_p / N) \times 100 \dots\dots\dots (2)$$

Where N_p = number of respondents who cited the use of a plant for a specific purpose, N = total number of respondents.

In this context, N_u refers to the total number of respondents who acknowledged all known medicinal uses of a particular species across various pharmacological contexts, including diarrhea treatment. Meanwhile, N_p indicates the number of respondents who specifically identified the use of that species for addressing a distinct diarrhea condition. A higher Fidelity Level (FL) value indicates stronger consensus among respondents on the species' targeted use, thereby highlighting its significant ethnomedical relevance for the specific ailment. These indices were used to determine the cultural significance and particular use of each medicinal plant cited by respondents.

Sample preparation

Fresh guava and betel leaves were washed, chopped, and dried at 40–50°C. The dried leaves were ground into a fine powder and sifted through a B-30 sieve. The powder was soaked in 70% methanol for 72 hours at room temperature, with occasional shaking. The liquid was then filtered and concentrated using a rotary evaporator at low pressure. The final extracts were stored at 4°C for later study (Farak et al., 2020). While the Osing tribe typically uses fresh or boiled water extracts, this study employed methanol to extract more bioactive compounds, such as flavonoids and alkaloids. Methanol improves extraction and stabilizes compounds for LC–MS analysis, enabling precise chemical profiling. Repeating LC–MS on methanol extracts was needed to confirm the specific metabolites in Osing plants, which might differ due to the area's unique environment.

Phytochemical screening

This research tested guava and betel leaf extracts to find the necessary plant chemicals. They used the methods described by Farak et al. (2020). To find alkaloids, they used Dragendorff's and Mayer's reagents. A reddish-brown solid indicated the presence of alkaloids. For flavonoids, they added magnesium powder and concentrated hydrochloric acid. A pink-to-red color indicated the presence of flavonoids. Tannins were found by adding 1% ferric chloride or 1% lead acetate. A dark green or yellow solid stated the presence of tannins. Saponins were tested by shaking the extract with hot water. Foam lasting more than 10 seconds meant saponins were present. Triterpenoids and steroids were found using the Liebermann–Burchard test. They added acetic anhydride and concentrated sulfuric acid. A reddish-purple color indicated the presence of triterpenoids, while green or blue stated the presence of steroids. These tests provided early chemical evidence for the traditional use of these plants in treating diarrhea.

Identification of phytochemical profile by LC-MS

This research used Liquid Chromatography–Mass Spectrometry (LC-MS) to identify specific compounds in guava and betel leaf extracts. They separated compounds from betel (*Piper betle*) and guava (*Psidium guajava*) leaves using an LC-MS machine. The process began by injecting the sample four times with a 5 µL microsyringe into a UPLC column (ACQUITY UPLC HClass System, Waters, USA) of type ACQUITY UPLC BEH C18 (2.1 × 50 mm). Samples were ionized using positive (+) ESI mode with a mass range of 50–1200 m/z. The machine was set to an ion source temperature of 100°C and a desolvation temperature of 350°C. Other settings included a cone gas flow rate of 0 L/h, a desolvation gas flow rate of 793 L/h, and a collision energy range of

4-60 eV. A gradient elution system was used with a flow rate of 0.2 mL/min. The peaks in the chromatogram were analyzed for data collection and study (Korfmacher, 2005).

Ligand preparation

We used 12 test compounds for ligand preparation. These compounds were chosen based on LC-MS profiling of *Psidium guajava* and *Piper betle* extracts. They represent the primary metabolites identified. The selection included quercetin, kaempferol, catechin, and piperine, known for their antidiarrheal, antioxidant, and anti-inflammatory effects. This method ensured the molecular docking analysis matched the chemical composition and therapeutic importance of the Osing tribe's medicinal plants. We drew the ligands using MarvinSketch software version 24.1.0. Then, we optimized their geometry and adjusted protonation at pH 7.4, saving the results in mrv format. After that, we performed a conformational search and saved the files in PDB and MOL2 formats.

Protein preparation and validation of docking method

The antidiarrheal receptor was obtained from the Protein Data Bank (PDB) at <https://www.rcsb.org/pdb/home/home.do>. Docking was performed using Autodock. Results with low binding affinity values were chosen and saved in PDB format (Ihsan, 2024). The docking method was verified by re-docking six proteins without ligands with their original ligands, using the AutodocTools 1.5.7 program (Ruswanto et al., 2020). Six protein targets were chosen because they are essential for understanding diarrhea and have good structural data in the PDB. These proteins are Cytochrome P450 1A1 (4I8V), Ephrin Type-B Receptor 4 (2VWY), Multidrug Resistance Protein 1 (7A69), Dihydrofolate Reductase (5HT4), Extracellular Calcium-Sensing Receptor (5FBH), and Rap Guanine Nucleotide Exchange Factor 4 (4MGI). Each receptor is linked to intestinal movement, secretion, or inflammation, which helps assess the antidiarrheal potential of the compounds. We used 100 GA Runs to improve accuracy. The only Genetic Algorithm parameter set was the number of GA runs (100×) to find the best position at which the receptor and ligand bind (Kelutur et al., 2020).

Analysis of docking results

From running, the results are obtained in the form of Binding energy and RMSD values. A good RMSD value is <2 Å. A binding energy value is considered good if it is lower. Receptors and ligands that have been docked are converted to PDB format and analyzed using Molegro Molecular Viewer (MMV) software, and the interactions are visualized in 2D and 3D (Ihsan et al., 2018).

Screening ligand-based drug likeness

Drug screening is performed on ligands with low binding energy and strong interactions with the target enzyme. Drug observation analysis is carried out with respect to the rule of good medicine (Lipinski's rule of five) (Ihsan et al., 2018).

RESULT AND DISCUSSION

Demographic characteristics of respondents

This ethnopharmacological study documents the traditional use of medicinal plants by the Osing people in Banjar and Licin village, Licin Subdistrict, Banyuwangi Regency, with a specific focus on antidiarrheal treatments. The selection of this site was strategic due to its high representation of Osing communities, which continue to preserve indigenous health practices. Based on Table 1, 34 respondents contributed valuable knowledge shaped by experience and cultural transmission. The gender composition of the respondents was relatively balanced, with a slight predominance of males (52.94%) over females (47.06%). This near-equitable distribution contrasts with findings from numerous ethnomedicinal investigations conducted in other parts of Indonesia, where male respondents typically predominate due to prevailing sociocultural structures (Febriyanti et al., 2024; Zubaidah et al., 2020). The relatively even gender representation observed in this study may reflect a more collaborative role between Osing men and women in the preservation, practice, and transmission of ethnomedicinal knowledge.

The respondents represented a broad age spectrum, with the majority falling within the 21 to 60-year range. Specifically, 17.65% were aged 21–30, 20.59% were aged 31–40, while 39.85% and 23.53% were in the 41–50 and 51–60 age brackets, respectively. Importantly, individuals aged 61–80 also made substantial contributions to the body of ethnomedicinal knowledge, underscoring the continued relevance of traditional practices among older community members. However, the notably low representation of respondents aged 80 or older (2.94%) raises concerns about the intergenerational transmission of this knowledge. Without deliberate efforts to transfer such expertise to younger generations, the gradual loss of elder knowledge holders may pose a significant risk to the preservation of ethnomedicinal heritage (Ouma, 2022).

Regarding educational attainment, half of the respondents (50.00%) had completed only elementary education. Other participants reported having completed junior high school (5.88%), senior high school (26.47%), or university education (5.88%), while 11.76% of respondents were illiterate. These findings suggest that traditional medicinal knowledge is not limited to older or non-literate individuals but is maintained across diverse educational backgrounds, highlighting the coexistence of formal and informal knowledge systems within the community (Oktavia et al., 2022).

Table 1. Demographic Characteristics of the Respondents participating in the ethnomedicinal survey

Distribution	Categories	Informant numbers	Percentage (%)
Genders	Male	18	52.94
	Female	16	47.06
Ages	21 until 30 years old	6	17.65
	31 until 40 years old	7	20.59
	41 until 50 years old	8	23.53
	51 until 60 years old	6	17.65
	61 until 70 years old	5	14.71
	71 until 80 years old	1	2.94
	81 until 90 years old	1	2.94
Education level	No formal education or Illiterate	4	11.76
	Elementary school	17	50.00
	Junior high school	2	5.88
	Senior high school	9	26.47
	University	2	5.88

Diversity of medicinal plant species used in the treatment of diarrhea

This study identified 9 medicinal plant species traditionally used by the Osing community for the treatment of inflammation and pain. Comprehensive ethnomedicinal data of these species are systematically organized and presented in Table 2. The table includes essential ethnobotanical indicators, such as Use Value and Fidelity Level, along with each plant's scientific name, botanical family, local designation, traditional therapeutic uses, specific plant parts used, preparation techniques, and modes of administration. The documented diversity reflects the depth of Osing ethnopharmacological knowledge and emphasizes the community's extensive reliance on plant-based remedies for managing inflammatory disorders and pain-related health issues.

According to Table 2, seven botanical families were reported to be used by the Osing community for the treatment of diarrhea. Among these, Musaceae and Rosaceae were the most frequently represented, each comprising two species. These were followed by Myrtaceae, Lauraceae, Zingiberaceae, Piperaceae, and Acoraceae, each represented by a single species. As

depicted in Figure 2.A, different plant parts were reported to be used by the Osing community to manage diarrhea. Leaves were the most commonly used part, with 32 instances (94.12%), followed by fruits, cited in 13 cases (38.24%), and rhizomes, which were mentioned in 4 cases (11.76%). The widespread use of leaves is consistent with previous ethnomedicinal research, which suggests that leaves are often preferred for their accessibility, renewability, and high levels of bioactive secondary metabolites such as flavonoids, alkaloids, and tannins (Bhagawan et al., 2022).

The variety of preparation techniques reflects the community's extensive ethnopharmacological knowledge, which is tailored to the specific therapeutic uses of each plant species (Figure 2.B). Regarding routes of administration (Figure 2.C), the majority of remedies were delivered orally (94%), highlighting the systemic approach commonly employed in traditional treatments for internal inflammatory ailments. In contrast, topical applications were documented in eight instances and were generally used to alleviate localized symptoms, such as bloating.

Based on ethnopharmacological interviews, nine plant species were identified as being used by the Osing people for the treatment of diarrhea. Table 2 lists these plants along with their species and botanical families. Guava (*Psidium guajava*) and betel (*Piper betle*) were cited most frequently.

Table 2. Medicinal plants used by the Osing tribe for diarrhea treatment

Familia Species	Name of Plant	Part used	Preparation	Adminis- tration	UV	FL
<u>Myrtaceae</u>						
<i>Psidium guajava</i>	Guava	Leaf	Raw, Boiled	Orally	2.44	64.7%
<u>Piperaceae</u>						
<i>Piper betle</i>	Betel	Leaf	Decoction	Orally, topically	1.11	29.4%
<u>Musaceae</u>						
<i>Musa paradisiaca</i>	Klutuk banana	Fruit	Raw	Orally	0.60	17.6%
<i>Musa sapientum</i>	Plantain	Fruit	Raw	Orally	0.33	8.82%
<u>Zingiberaceae</u>						
<i>Curcuma longa</i>	Turmeric	Rhizome	Decoction	Orally	0.44	11.7%
<u>Rosaceae</u>						
<i>Rubus rosa</i>	Calingan	Leaf, fruit	Raw	Orally	0.33	8.82%
<i>Potentilla arguta</i>	Grunggung	Leaf	Crush, boiled	Orally, topically	0.11	2.94%
<u>Acoraceae</u>						
<i>Acorus calamus</i>	Dringu	Leaf, Fruit	Crush	Topically	0.11	2.94%
<u>Lauraceae</u>						
<i>Litsea cubeba</i>	Kranggean	Leaf, Fruit	Decoction	Orally	0.11	2.94%

The study utilized quantitative ethnobotanical indices—Use Value (UV) and Fidelity Level (FL)—to evaluate the relevance and specificity of medicinal plant use. The UV represents the importance of each plant species based on the number of mentions across respondents. To assess the significance of each plant, a Use Value (UV) was calculated for each species (Table 2). Guava recorded the highest UV (2.44), followed by betel (1.11). These findings suggest that guava and betel are not only frequently cited but also considered adequate by the Osing community. A similar pattern was observed by Mustofa (2018), who reported UV and FL values of 3.44 and 2.08, respectively, in plants used by traditional healers in Sulawesi (Zubaidah et al., 2020).

The findings align with earlier studies in other Osing-populated areas, such as Singojuruh and Kabat Subdistricts, which also reported the use of *Psidium guajava* (guava) leaves to treat diarrhea. These consistencies underscore the significance of local plant knowledge as a cultural and therapeutic resource (Zaman et al., 2021).

Fidelity level of medicinal plant species

Table 2 presents the medicinal plants identified through Fidelity Level (FL) analysis as being used by the Tengger community for the treatment of inflammation and pain. The FL values of the documented species ranged from 1.11% to 100%, reflecting varying levels of consensus among respondents regarding their specific therapeutic applications. *Psidium guajava* demonstrated the highest FL for diarrhea (FL: 64.74%), followed by *Piper betle*, which also showed a relatively high FL value (FL: 29.4%) for the same condition. The prominence of these plants in addressing diarrhea within the study area is likely influenced by their widespread use in traditional Indonesian medicine for managing various health disorders (Arozal et al., 2020).

These values reflect both the cultural significance and therapeutic reputation of these plants. Ethnopharmacology not only explores the medicinal potential of plant species but also considers the socio-cultural context in which this knowledge is practiced and transmitted (Laily et al., 2015). The Osing people's reliance on guava and betel illustrates a well-established empirical tradition, grounded in generations of observation and experience (Bhagawan et al., 2022).

The phytochemical screening of guava and betel leaf extracts supported the ethnobotanical data. Both were found to contain alkaloids, flavonoids, tannins, and steroids, all of which are known for their pharmacological activities (Amat-Ur-Rasool et al., 2020). Flavonoids and tannins, in particular, are widely recognized for their antidiarrheal properties due to their astringent and anti-secretory effects. Previous studies also confirm the antioxidant properties of guava and betel leaves, further enhancing their therapeutic potential (Kumar et al., 2021a). In Table 3, phytochemical screening of guava and betel leaf extracts revealed the presence of alkaloids, flavonoids, tannins, and steroids in both samples, while saponins were not detected.

Table 3. Phytochemical constituents of guava and betel leaf extracts

Phytochemical Test	Reagent	Result	
		Guava Leaf	Betel Leaf
Alkaloid	Mayer	+	+
	Bouchardat	+	+
	Dragendorff	+	+
Flavonoid	powder of Mg, add some amyl alcohol	+	+
	HCl in concentrated	+	+
Tannin	FeCl ₃	+	+
Saponin	Shake hot water vertically	-	-
Triterpenoid or Steroid	Liebermann - Bouchardat	+	+

Note: “+” indicates presence; “-” indicates absence

To support and verify the preliminary findings, LC-MS analysis was performed to detect specific secondary metabolites. The results corroborated the outcomes of the phytochemical screening, notably by identifying flavonoid compounds, including quercetin and kaempferol (Garrido et al., 2024). These bioactive constituents are well documented in the scientific literature for their therapeutic roles in gastrointestinal health. Further LC-MS analysis supported the phytochemical screening findings by detecting specific secondary metabolites. The chromatograms showed firm peaks for quercetin and kaempferol in both guava and betel leaf extracts (Figure 2). To further validate these findings, LC-MS analysis was conducted to identify specific secondary metabolites. The results are in Table 4. The results were consistent with the phytochemical screening, particularly in the identification of flavonoids such as quercetin and kaempferol (Kumar et al., 2021b). These compounds have been widely studied for their effects on the gastrointestinal tract. Quercetin is known to reduce intestinal motility and secretion, while kaempferol has demonstrated anti-inflammatory and spasmolytic activity (Ahmad et al., 2023). According to Faulinza (2022), guava leaves with high quercetin content significantly reduced the incidence of diarrhea. Kurnia et al. emphasized the importance of further exploring quercetin's potential using

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chemotaxonomic approaches. The identification of these compounds in both guava and betel supports the hypothesis that these plants have pharmacological efficacy against diarrheal disorders.

This mix of old knowledge and new science shows how ethnopharmacology can connect cultural practices with proven medicine. The findings highlight the need to preserve local knowledge, which is culturally valuable and may help identify new treatments. However, this study has limits. The small sample size and lack of lab tests make it hard to draw firm conclusions about effectiveness. Future research should include lab tests, safety checks, and possibly studies on how to make these plant remedies to determine their dosages, safety, and effectiveness.

In summary, the Osing community uses guava and betel leaves to treat diarrhea. This practice is part of their culture and has scientific support. The plants have high UV and FL values and contain essential compounds that help with diarrhea. This shows the importance of protecting traditional knowledge and suggests the need for further research. *Piper betle* and *Psidium guajava* are necessary in modern studies. A study found 14 active components in them. These findings show that *Piper betle* and *Psidium guajava* have potential as natural medicines. They have been used in traditional medicine for various health issues. Flavonoids in these plants have strong antioxidant properties (V et al., 2025). Alkaloids and phenolic compounds exhibit antimicrobial activity, and terpenoids such as andrographolide may protect the liver. The complex mix of these compounds suggests many health benefits, making these plants good candidates for natural drug development (Garrido et al., 2024).

The LC–MS analysis of guava and betel leaf extracts found several active compounds, mainly flavonoids, phenolic acids, and alkaloids. These compounds are known for their antidiarrheal, antioxidant, and antimicrobial effects, which match the Osing tribe's use of these plants for stomach issues. The study confirmed that quercetin and kaempferol are significant components in both extracts, supporting their medical importance.

Table 4. Identified Chemical Compounds from LC–MS Analysis of Guava (*Psidium guajava*) and Betel (*Piper betle*) Leaf Extracts

Compound Name	Molecular Formula	Retention Time (min)	Molecular Weight (g/mol)	Detected In	Reported Pharmacological Activity
Quercetin	C ₁₅ H ₁₀ O ₇	4.65	302.24	Guava, Betel	Antidiarrheal, anti-inflammatory, antioxidant
Kaempferol	C ₁₅ H ₁₀ O ₆	9.30	286.23	Guava, Betel	Spasmolytic, antimicrobial, antioxidant
Catechin	C ₁₅ H ₁₄ O ₆	2.49	290.27	Guava, Betel	Antioxidant, intestinal motility regulator
Apigenin	C ₁₅ H ₁₀ O ₅	7.84	270.23	Guava	Anti-inflammatory, smooth muscle relaxant
Ferulic acid	C ₁₀ H ₁₀ O ₄	10.65	194.18	Guava	Antioxidant, antidiarrheal
Caffeic acid	C ₉ H ₈ O ₄	4.20	180.15	Guava	Antimicrobial, anti-secretory
Isorhamnetin	C ₁₆ H ₁₂ O ₇	8.21	316.26	Guava	Anti-inflammatory, cytoprotective
Piperine	C ₁₇ H ₁₉ NO ₃	13.48	285.33	Betel	Antimicrobial, anti-spasmodic
Eugenol	C ₁₀ H ₁₂ O ₂	0.57	164.20	Betel	Antibacterial, analgesic
Safrol	C ₁₀ H ₁₀ O ₂	7.70	162.18	Betel	Antimicrobial, antioxidant
Myristicin	C ₁₁ H ₁₂ O ₃	5.76	192.21	Betel	Anti-inflammatory, smooth muscle relaxant
Gallocatechin	C ₁₅ H ₁₄ O ₇	3.62	306.26	Guava, Betel	Antioxidant, anti-secretory
Rutin	C ₂₇ H ₃₀ O ₁₆	6.73	610.51	Guava	Antioxidant, relaxant
Epicatechin	C ₁₅ H ₁₄ O ₆	2.97	290.26	Guava, Betel	Antioxidant, anti-secretory

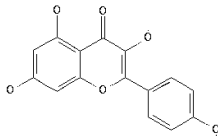
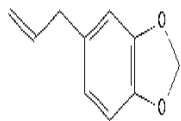
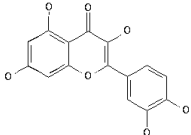
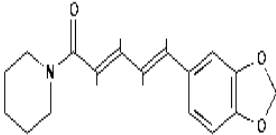
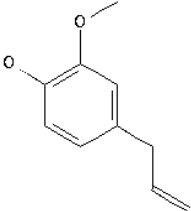
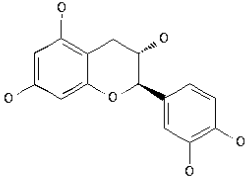
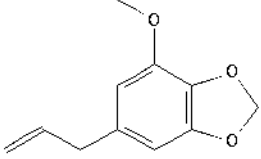
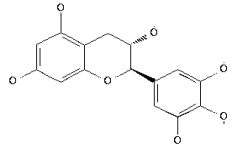
Ligand preparation

Molecular docking studies how two or more molecules, like a drug and a protein, fit together. It predicts how a protein interacts with a small molecule. The process starts by preparing the test molecule. In this study, compounds from guava and betel leaves are used. Table 5 is performed on 14 compounds to examine their interactions with 6 receptors. The structure of each compound is opened in MarvinSketch, cleaned, protonated in 2D, and saved in MRV format. The file is reopened to confirm, and the lowest bond energy value is selected and saved in the MOLE 2 format. The goal of the simulation is to identify the most effective molecule by assessing its ability to bind to the receptor.

Table 5. Structures of test ligands from guava leaves (*Psidium guajava* L)

Compound	2D Structure	Compound	2D Structure
Kaempferol Structure: C ₁₅ H ₁₀ O ₆ Retention time (minutes): 9.3 Molecular weight (g/mol): 286.23		Epicatechin Structure: C ₁₅ H ₁₄ O ₆ Retention time (minutes): 2.97 Molecular weight (g/mol): 290.26	
Quercetin Structure: C ₂₁ H ₂₀ O ₁₂ Retention time (minutes): 4.65 Molecular weight (g/mol): 464.37		Gallocatechin Structure: C ₁₅ H ₁₄ O ₇ Retention time (minutes): 3.62 Molecular weight (g/mol): 306.26	
Apigenin Structure: C ₁₅ H ₁₀ O ₅ Retention time (minutes): 7.84 Molecular weight (g/mol): 270.23		Isorhamnetin Structure: C ₁₆ H ₁₂ O ₇ Retention time (minutes): 8.21 Molecular weight (g/mol): 316.26	
Ferulic acid Structure: C ₁₀ H ₁₀ O ₄ Retention time (minutes): 10.65 Molecular weight (g/mol): 194.18		Rutin Structure: C ₂₇ H ₃₀ O ₁₆ Retention time (minutes): 6.73 Molecular weight (g/mol): 610.51	
Caffeic Acid Structure: C ₉ H ₈ O ₄ Retention time (minutes): 4.2 Molecular weight (g/mol): 180.15		Cathechin Structure: C ₁₅ H ₁₄ O ₆ Retention time (minutes): 2.49 Molecular weight (g/mol): 290.46	

Table 6. Structure of test ligands from betel leaf (*Piper betle*).

Compound	2D Structure	Compound	2D Structure
Kaempferol Structure: C ₁₅ H ₁₀ O ₆ Retention time (minutes): 6.17 Molecular weight (g/mol): 286.23		Saffrol Structure: C ₁₀ H ₁₀ O ₂ Retention time (minutes): 7.70 Molecular weight (g/mol): 162.18	
Quercetin Structure: C ₂₁ H ₂₀ O ₁₂ Retention time (minutes): 4.65 Molecular weight (g/mol): 464.37		Piperin Structure: C ₁₇ H ₁₉ NO ₃ Retention time (minutes): 13.48 Molecular weight (g/mol): 285.33	
Eugenol Structure: C ₁₀ H ₁₂ O ₂ Retention time (minutes): 0.57 Molecular weight (g/mol): 164.2		Cathecin Structure: C ₁₅ H ₁₄ O ₆ Retention time (minutes): 2.65 Molecular weight (g/mol): 290.46	
Myristicin Structure: C ₁₁ H ₁₂ O ₃ Retention time (minutes): 5.76 Molecular weight (g/mol): 192.21		Galocatechin Structure: C ₁₅ H ₁₄ O ₇ Retention time (minutes): 9.81 Molecular weight (g/mol): 306.26	

Protein preparation

First, we prepare the test ligand. Next, we prepare the receptor. Drug receptors are specific large molecules, like lipoproteins and nucleic acids. They bind to ligands such as drugs, hormones, and neurotransmitters. This binding transmits chemical signals within cells, resulting in effects (Salsabila et al., 2023). We downloaded the structure of these large molecules from the Protein Data Bank (PDB) website at <http://www.rcsb.org>. The PDB is a collection of data on the 3D structure of these molecules. We open the file with the MOLEGO molecular viewer software to separate the native ligand from the protein. We then save it in a working folder in PDB format. Next, we remove water molecules using Discovery Studio.

Removing water molecules helps reduce the computational load, making the simulation faster. After removing water, we add hydrogen atoms to restore the body's pH. In this study, we used six receptors: Cytochrome P450 1A1 (PDB code id: 4I8V), Ephrin Type-B Receptor 4 (PDB code id: 2VWY), Multidrug resistance protein 1 (PDB code id: 7A69), Dihydrofolate reductase (PDB code id: 5HT4), Extracellular calcium-sensing receptor (PDB code id: 5FBH), and Rap guanine nucleotide exchange factor 4 (PDB code id: 4MGI). These receptors are linked to important gut processes like secretion, inflammation, absorption, and movement. This helps us assess the antidiarrheal activity of compounds from *Psidium guajava* and *Piper betle*.

Table 7. List of native ligands for each receptor

Protein Target	PDB Code	Native Ligand	Molecular Formula	Function/ Binding Role
<i>Cytochrome P450 1A1</i>	4I8V	BHF (2-phenyl-4H-benzo[h]chromen-4-one)	C ₁₅ H ₁₀ O ₂	Oxidation and detoxification enzyme substrate
<i>Ephrin Type-B Receptor 4</i>	2VWY	7X5N (5-chloro-1,3-benzodioxol-4-yl)-N-(3-methylsulfonylphenyl)pyrimidine-2,4-diamine	C ₁₆ H ₁₃ ClN ₄ O ₄ S	Signal transduction, epithelial integrity
<i>Multidrug Resistance Protein 1</i>	7A69	R1Q (Vincristine)	C ₄₆ H ₅₆ N ₄ O ₁₀	Efflux pump modulator
<i>Dihydrofolate Reductase</i>	5HT4	65J (7-ethyl-6-[(3-methoxyphenyl)sulfonyl]-5-methyl-7H-pyrrolo[2,3-d]pyrimidine-2,4-diamine)	C ₁₆ H ₁₆ N ₆ O ₂ S	Antimicrobial/ folate pathway inhibitor
<i>Extracellular Calcium-Sensing Receptor</i>	5FBH	CSO (S-hydroxycysteine)	C ₃ H ₇ NO ₃ S	Regulates calcium and smooth muscle contraction
<i>Rap Guanine Nucleotide Exchange Factor 4</i>	4MGI	CMP (Cyclic adenosine monophosphate)	C ₁₀ H ₁₂ N ₅ O ₆ P	Controls intestinal secretion via cAMP signaling

The basis for the assessment is the Root Mean Square Deviation (RMSD) value. The method is considered valid if the RMSD is < 2 Å. The smaller the error in the calculation results, the more accurate the calculation; in addition, the lower the RMSD value, the more stable the protein (Kaharudin et al., 2022).

Table 8. Redocking results

Native ligand	Run	Binding energy (kcal/mol)	RMSD (Å)
BHF	77	-10.08	1.02
7X5N	30	-9.57	0.60
R1Q	11	-2.78	31.96*
CSO	35	-7.47	3.25*
65J	26	-2.53	1.83
CMP	35	-9.01	0.55

* does not meet the criteria

The table above shows the results of running on six native ligands. The above native ligand meets the validation criteria of the docking method, with an RMSD < 2 Å. However, docking validation was conducted by re-docking native ligands into their respective protein active sites, with RMSD values below 2.0 Å considered acceptable. Most proteins met this criterion, confirming the reliability of the docking. Those with RMSD values exceeding 2 Å (e.g., the R1Q–7A69 and CSO-5FBH complexes) were excluded from further analysis. This approach ensured that only structurally valid and reproducible docking models were used to interpret the interactions of using plant compounds.

Table 9 lists the test ligands that meet the Lipinski constraints. Furthermore, these ligands will undergo virtual screening to obtain the binding energy of each test ligand by running simulations between the ligand and prepared protein macromolecules. Binding energy is a measure

of a drug's affinity for a receptor. The smaller the binding energy value, the higher the affinity between receptors and ligands; conversely, if the binding energy value is greater, the affinity between receptors is lower (Shofi, 2021).

Table 9. Molecular tethering results of the native ligand and Lipinski results for each compound

Compound	Molecular Weight (<500g/mol)	Hydrogen Donor (<5)	Hydrogen Acceptor (<10)	Log p (<5)	Binding energy			
					BHF	7X5N	65J	CMP
Kaempferol	286.239	4	6	2.2824	-7.54	-7.18	-6.41	95.38
Quercetin	302.238	5	7	1.988	-9.72	-6.92	-6.87	150.38
Safrol	162.182	0	3	3.4559	-8.08	-6.64	-6.47	83.91
Isorhamnetin	316.269	7	8	0.7426	-3.65	-6.58	-5.64	194.49
Cathecin	290.271	5	6	1.5461	-7.69	-7.6	-7.23	113.27
Apigenin	270.230	3	3	-0.6719	-3.19	-2.58	-3.22	-1.46
Piperin	285.332	2	2	-0.8208	-3.36	-4.41	-4.22	-1.41
Caffeic acid	180.155	1	1	0.111	-4.23	-3.50	-5.87	186.5
Ferulic acid	224.211	1	1	0.4659	-2.7	-4.31	-2.44	-2.25
Epicatechin	290.266	0	2	2.2166	5.08	-4.20	-4.96	5.79
Gallocatechin	306.263	3	3	1.6916	6.88	-5.65	-6.05	31.4
Myristicin	192.211	2	5	2.3767	-9.4	-7.37	-5.48	199.33
Rutin	610.512*	3	3	0.211	does not meet the criteria			
Eugenol	164.211	2	5	0.4759	-4.41	-3.75	-3.73	14.68

Quercetin is the best test compound because it binds well to several receptors with the lowest binding energy. In this study, quercetin's binding energy is -9.72 kcal/mol, which is stronger than in other studies on *Ruellia tuberosa* (-8.36 kcal/mol) in Shofi (2021) and *Melastoma malabathricum* (-8.45 kcal/mol) in Salsabila (2023). This suggests that plants from the Osing region have a higher receptor affinity. The difference might be due to the unique environment and plant chemicals in the Ijen Highlands. Earlier studies focused on single targets, whereas this study examined six proteins involved in the digestive system. This demonstrates quercetin's potential as a medicinal compound and that Osing plant extracts have strong and varied interactions with receptors. This supports the traditional use of Osing medicine.

CONCLUSION

This study examined how the Osing community in Licin and Banjar Village utilizes guava and betel leaves to treat diarrhea. These plants are the most commonly used for this purpose. Tests have shown that both guava and betel leaves contain high amounts of quercetin and kaempferol. These are compounds that help with stomach issues. Among 14 compounds tested in these leaves, quercetin showed the strongest binding to specific receptors, making it a good candidate for an anti-diarrheal drug. However, further studies are needed to confirm the effectiveness and safety of these plants for medicinal use.

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